

Watch when no one's watching



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In these days of rising crime rates, surveillance assures us of some safety. The sole purpose of this set up is to be able to monitor your home while you are away. We advise you to maintain restraint and use this set up only for ethical purposes. Whether you have young children at home, or whether you're on vacation wondering if your home is safe, this short guide should help you without burning a hole in your pocket.

Web cams are no longer luxury. Socialising by means of IMs, video-conferencing through clients such as Skype and the likes have made web cams a part of today's standard PC configurations. After completing this workshop, you should be able to view the latest image from your home in real time or view sequential images archived on a PC that could later serve as evidence in the unfortunate event of a crime.

Your set up largely depends on how far you can stretch your budget and the number of rooms you need to monitor. You could either use a single-cam or multi-cam configuration. You also have the option of implementing network web cams into your design. This way you don't need a PC powered on all the time.

Nevertheless, the cost of such a set up is atrocious. For a rough estimate, a basic IP web cam costs approximately Rs. 8,000; while you would get a decent web cam starting at Rs. 800. The downside

in this method, however, is that your PC would need to be powered on as long as you need surveillance.

In addition to the web cams, you would need web cam software to process the still images and direct the feed of the web cam to appropriate folders. There is a long list of software available online. One such software suitable for this application is Dorgem. At 400 KB, it's a relatively small executable that has some promising surveillance features.

You can download dorgem at <http://dorgem.sourceforge.net/download.html>. You can also find it on our DVD. The initial set up doesn't take too long and in seconds you should be able to view images from your web cam. The best feature of Dorgem is the ability to send your feeds either via HTTP, FTP, save them as files or even let an external program further modify them.

The Dorgem interface is basically divided into two sections. You have Camera options on the top, followed by Caption and Storage settings. First, select your web cam from



Configuring the camera just takes a minute

the drop down in the middle. For typical plug and play devices, select Microsoft WDM Image Capture (Win32). Click on the Source button.

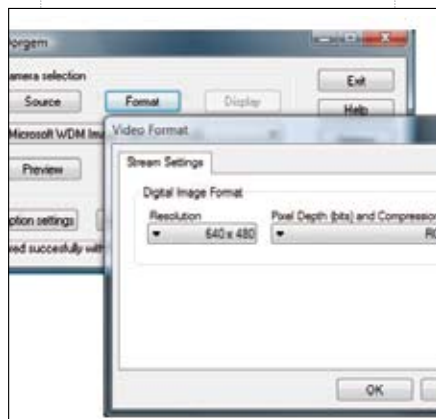
Under the Capture Source tab, select your device. If you're using your laptop, select Integrated web cam, else select your camera by name. You can also set various parameters for the desired image. For example, if you are going to monitor a darker room or under low-light conditions, then increase the brightness levels and change the contrast levels accordingly. Also, if the entry/exit points are far from the position of the camera, adjust the zoom level. Check the visible feed during initial set up. If you need an archived record of all movements happening in your home, you can use the wildcard %g after the filename. This way, each image generated after the stipulated time is named

with the timestamp and saved on the location specified. You can set an appropriate capture interval as 5 sec.

Depending on the available bandwidth or speed of your system, you can select the frame size of the image. We recommend a frame size of 320x240. This way, the file size of each image is relatively small; as a result, the system response is quicker and files are dumped faster – be it on the PC or on a remote

server. In order to view your files from a remote location, you simply need to configure an FTP server on the PC to which your web cams are connected.

For this, you need FileZilla FTP Server. The whole process can be further simplified if you have access to file hosting services. Simply enter the database username and password and set a location. The images would then be uploaded to your web site. This way you could monitor it from any PC that can connect to the internet. Alternatively, you can also view the image by remotely accessing the PC on which the images are saved. For this, type the command `ftp://<username>:<password>@<ip add or domain>`. Here, username and password are those set on the FileZilla server on the host PC. Depending on the speed of your connection (especially on slow systems and connections), adjust the image size and intervals to avoid conflicts.



Adjust the image settings for optimum results, especially file size